

2-5 狄拉克符号

1 2-1 旋转矩阵

$R(\theta)$ 在基 $\{|0\rangle, |1\rangle\}$ 下的矩阵元为 $R_{ij} = \langle i|R|j\rangle$, 按外积展开:

$$\begin{aligned} R(\theta) &= \cos \theta (|0\rangle\langle 0| + |1\rangle\langle 1|) + \sin \theta (|0\rangle\langle 1| - |1\rangle\langle 0|) \\ &= \cos \theta I + \sin \theta (|0\rangle\langle 1| - |1\rangle\langle 0|). \end{aligned}$$

作用在初态 $|0\rangle$ 上:

$$R(\theta)|0\rangle = \cos \theta |0\rangle - \sin \theta |1\rangle. \quad (1)$$

2 2-2 矩阵指数

三个泡利矩阵的外积 (Dirac) 形式为

$$\sigma_x = |0\rangle\langle 1| + |1\rangle\langle 0|, \quad (2)$$

$$\sigma_y = -i|0\rangle\langle 1| + i|1\rangle\langle 0|, \quad (3)$$

$$\sigma_z = |0\rangle\langle 0| - |1\rangle\langle 1|. \quad (4)$$

由 $\hat{P}^2 = I$ (见 2-2 证明), 矩阵欧拉公式

$$e^{i\theta\hat{P}} = \cos \theta I + i \sin \theta \hat{P}, \quad I = |0\rangle\langle 0| + |1\rangle\langle 1|. \quad (5)$$

3 2-3 期望

矩阵 $\hat{Q}$ 关于列向量 v 的期望, 即 bra-ket 形式

$$v^\dagger \hat{Q} v = \langle v | \hat{Q} | v \rangle. \quad (6)$$

取 $v(\theta) = e^{i\theta\hat{P}/2}|0\rangle$, 注意到 $\hat{P}$ 厄米, 故 $(e^{i\theta\hat{P}/2})^\dagger = e^{-i\theta\hat{P}/2}$, 期望为

$$\langle \hat{Q} \rangle = \langle 0 | e^{-i\theta\hat{P}/2} \hat{Q} e^{i\theta\hat{P}/2} | 0 \rangle. \quad (7)$$

4 2-4 张量积

n 比特计算基是单比特基的张量积:

$$|q_0 q_1 \cdots q_{n-1}\rangle = |q_0\rangle \otimes |q_1\rangle \otimes \cdots \otimes |q_{n-1}\rangle. \quad (8)$$

单点与双点算符（下标为位置，从左数 0 起点）：

$$Z_i = I^{\otimes i} \otimes \sigma_z \otimes I^{\otimes (n-i-1)}, \quad (9)$$

$$X_i X_{i+1} = I^{\otimes i} \otimes \sigma_x \otimes \sigma_x \otimes I^{\otimes (n-i-2)}. \quad (10)$$

哈密顿量及其在 $|0\rangle^{\otimes n} = |00\cdots 0\rangle$ 下的期望：

$$H = \sum_{i=0}^{n-1} Z_i + \sum_{i=0}^{n-2} X_i X_{i+1}, \quad \langle 00\cdots 0 | H | 00\cdots 0 \rangle = n. \quad (11)$$

（因 $\langle 0 | \sigma_z | 0 \rangle = 1$ 贡献 n ， $\langle 0 | \sigma_x | 0 \rangle = 0$ 贡献 0。）

5 目标列向量

$$\frac{1}{\sqrt{2}}(|010\rangle - |101\rangle) = \frac{1}{\sqrt{2}}(|0\rangle \otimes |1\rangle \otimes |0\rangle - |1\rangle \otimes |0\rangle \otimes |1\rangle) \quad (12)$$

$$= \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ -1 \\ 0 \\ 0 \end{pmatrix} \quad (13)$$